Supervised Fine Tuning

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The Base Model

- **Core Objective**: Pre-trained for Next Token Prediction (Causal Language Modeling)
- **Type of Learning**: Self-Supervised Learning
- **Training Data**: Massive, uncurated datasets representing the "statistical distribution of the internet" (Common Crawl, Wikipedia, GitHub)
- **Probability Engine**: Calculates the likelihood of the next word based on previous context ($P(W_t/W_{1:t-1})$)

The Base Model

- **Behavioral Identity**: It is a text completer.
- **In-Context Learning**: They can mimic a pattern if provided examples.
- It is not explicitly trained to follow instructions.
- **Lack of Dialogue State**: No inherent concept of “User” vs “Assistant”. It treats everything as a single, continuous stream of text.

Completion Trap

- **Statistical Mimicry:** As the base model is a text completion engine, it prioritizes ‘**style and format**’ of the input over the ‘**intent**’ of the instruction.
- **Lack of State switching**: No concept of “turn-taking”. Model treats the prompt as first half of a single, continuous document.
- **Probability vs Utility**: The most probable solution is not the most useful one.

Supervised Fine Tuning

- Moving to goal-oriented Instruction Following.
- Step 1: Create a dataset of (Instruction, Response) pair.
- Step 2: Train the Model on “Gold Standard” response Y given an instruction X.
- **Supervised Learning**: Unlike the curated pre-training data, SFT uses a high-quality, human-labeled dataset of (Instruction, Response) pairs.

Supervised Fine Tuning

- **Role Identification**: Introduction of special tokens to define boundaries between the User (the environment) and the Assistant(the agent).
- **Format**: <|im_start|>user\n{Instruction}<|im_end|>
- <|im_start|>assistant\n{Response}<|im_end|>
- **Intent Alignment**: The gold standard response are created in such a way that the model prioritizes being “Helpful, Honest and Harmless” over mere statistical probability.

Concept of Alignment

- Ensuring model’s behavior matches human intent, values, and specific task requirements.
- The “Manifold” metaphor: SFT carves a narrow “Instruction-Following Manifold” within massive high-dimensional space of base model.
- **Prompting as steering**: Prompt keeps the model on the aligned model and prevents “drift” back into base completion mode.
- **Prompt Engineering** provides the context require to trigger the correct capability.

Anatomy of Gold Sample

- **Triplet Structure** (Instruction, Input, Response)
- **Instruction**: Specific command or task (e.g. “Summarize this)
- **Input** (Optional): The data or context to be processed
- **Response**: The “Gold Standard” output that the model must learn to emulate.
- **“Gold Standard”**: Samples created by human experts or high-capability “Teacher” models to ensure accuracy, tone and formatting.

Anatomy of Gold Sample

- **Data format**: Typically stored in structured formats like JSONL
- **Selective loss calculation:** Loss functions is only applied to tokens in response section.
- **Delimiters and Metadata**: Samples use unique separators (e.g. ### Instruction:, ### Input:, ### Response:) to define clear boundaries between task and data.
- A good prompt is essentially **an attempt to recreate this specific triplet structure at inference time**.

Direct Instruction Samples

- Training pairs consisting of a **simple command** and its **direct execution**.
- Example: Instruction: “Translate ‘Hello’ to French.” → Response: “Bonjour.”
- **Verb to task mapping**: Teaches the model that specific imperative verbs (Summarize, Extract, Rewrite, Classify) map to specific computational transformations.
- **Standardized formatting**: Turning a paragraph into a list or table.

Chain-of-Thought (CoT) Traces

- Training samples where the response includes the step-by-step logical derivation before the final result.
- **Structure**: [Complex Problem] → [Reasoning/Logic Path] + [Final Answer]
- **Token-Based Compute**: Teaches the model to use intermediate tokens as memory to resolve complex dependencies before committing to a final output.
- **Triggering the weights**: A prompt with above structure makes the the model trigger the weights corresponding to reasoning/logic path.

MCP-Driven Trajectories

- Model Context Protocol (MCP): Communication protocol for LLM to interact with external tools (databases, file systems, APIs)
- Recording Expert Trajectories: A high-capability “Teacher” model uses real MCP solvers to solve a task, recording every step of the tool interaction as a gold sample. This is known as Trajectory SFT Sample.
- Teach the model to query external tool and understand from responses.

MCP-Driven Trajectories

The Anatomy of an MCP Trace:

- **Thought**: The model's internal plan/reasoning for using a tool.
- **Tool Call**: The standardized MCP-compliant request (e.g., a specific JSON schema).
- **Observation**: The actual data returned from the MCP server (the "Real World" feedback).
- **Synthesis**: The final reasoning step incorporating the tool's output.

Agent

- The model is given a **Persona**.
- Model knows how to **respond to instructions**.
- The model knows how to **respond by following a chain of thought**.
- The model knows how to **call APIs** for reading/writing databases, web crawling etc.
- It is an **Agent**.

Types of AI Agents

- **Instruction-Following Agent**: Given an instruction, it prints a response. It is primarily reactive.
- **Reasoning Agent (ReAct)**: Uses CoT to verbalize a plan before executing action.
- **Tool-Augmented agents**: Uses Standardized protocols (MCP) to interface with external environments.

Types of AI Agents

- **Autonomous Agents**: Work with high level objectives (eg. Stock analyst) and manage their own sub-goals and task prioritization.
- **Reflexive/Self-correction Agents**: Specifically trained on “Critic” trajectories. They generate an initial output, critique it against a rubric and iterate until the goal is met.
- **Multi-Agent System**: Many agents working together.